

Highly regarded for his groundbreaking research in brain imaging, Portuguese neurologist António Egas Moniz also developed the controversial surgical operation that became known as the frontal lobotomy. His technique, refined by other neurologists, became a central part of psychiatric treatment in the 1940s and 1950s.

In 1927, António Egas Moniz, the University of Lisbon's first neurological professor, invented the angiogram, a type of X-ray for checking blood vessels. The invention was driven by Moniz's interest in how to better identify the position of brain tumors to improve the chances of their successful removal. He believed that if the blood vessels of the brain could be seen by radiography, tumors could be precisely located. Experimenting on human subjects with injections of radiopaque dyes, he found that a solution of 25 percent sodium iodine gave the safest and clearest result. Moniz's technique became the only diagnostic tool used to image cerebral vessels and identify blocked arteries until the advent of CT imaging in 1975.

Moniz was a pioneer in psychosurgery, too. He believed that he could eradicate certain mental states, such as obsession and depression, by operating on the frontal lobes of mentally ill patients. He designed a needle with a wire loop, a leucotome, for cutting through but not removing the connecting tissue of the lobes. His lobotomies (also known as leucotomies) on patients suffering from depression, anxiety, and schizophrenia were thought to alter associated personality traits and behaviors. The technique, further developed in the US, became a standard procedure for treating the mentally ill until the 1960s. However, only a third of patients benefited, and others experienced serious harm.

*Before the invention of brain scanners, angiograms—invented by Moniz—were one of the best ways to study abnormalities inside the brain.*

“His life was **unusually productive**; his name will live for his **two great contributions ...**”

Geoffrey Jefferson, 1955

#### MILESTONES

##### VASCULAR IMAGES

Begins experimentation on vascular imaging with the help of Pedro Almeida in 1902.

##### ANGIOGRAPHIC PAPER

Publishes his first paper on angiography in the French journal *Revue Neurologique* in 1927.

##### NEAR-FATAL SHOOTING

Shot by a schizophrenic patient in 1939. His spine is shattered, but he continues to practice.

##### NOBEL AWARD

Receives Nobel Prize in Physiology or Medicine in 1949 for invention of prefrontal lobotomy.

